# Scalability & Future Plan

| | |
|---|---|
| **Date** | 2 July 2026 |
| **Team ID** | XXXXXX |
| **Project Name** | OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE |
| **Maximum Marks** | 1 Mark |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation | Impact | Priority to Address (High / Medium / Low) |
|---|---|---|---|
| 1 | System currently runs on a local Flask server | Limited accessibility for multiple users | High |
| 2 | Uses a static CSV dataset | Dataset updates require manual intervention | Medium |
| 3 | No real-time weather integration | Predictions rely only on user-entered values | High |

### Scalability Plan:

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | User Load | Supports limited users on localhost | Deploy on cloud platform to support thousands of users |
| 2 | Data Storage | CSV-based dataset storage | Use MySQL/PostgreSQL database for efficient storage and retrieval |
| 3 | Performance | Local model prediction | Optimize model and use cloud infrastructure for faster predictions |
| 4 | Security | Basic application security | Implement authentication, HTTPS, and secure data storage |

**Future Roadmap:**

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|
| Phase 1 | Deploy application on cloud platform | 3 Months | Improved accessibility and scalability |
| Phase 2 | Real-time weather API integration | 6 Months | More accurate crop recommendations |
| Phase 3 | Fertilizer recommendation system | 9 Months | Better crop productivity and soil management |
| Phase 4 | Mobile application for farmers | 12 Months | Increased usability and wider adoption |